

# AI-Powered Product Intelligence System

## Student Information

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**Assignment:** AI-Powered Product Intelligence System

**Tools Used:** Python, BLIP, CLIP, FAISS, Scikit-Learn, Kaggle Notebook

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## 1. Introduction

Modern e-commerce platforms contain thousands of products, making it difficult for customers to discover relevant items and for businesses to maintain a clean and organized catalog. Artificial Intelligence can help solve these challenges by automatically understanding product images, generating meaningful representations, and enabling intelligent search and recommendation systems.

In this project, an AI-Powered Fashion Product Intelligence System was developed using multimodal machine learning models. The system builds upon the instructor's implementation that uses BLIP for image understanding, CLIP for multimodal embeddings, and FAISS for similarity search. The project further extends the base system by implementing three advanced features:

1. Outfit Recommendation System
2. Duplicate Product Detection
3. Natural Language Product Search

The final solution demonstrates how a single embedding pipeline can support multiple real-world e-commerce applications.

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## 2. Problem Statement

During the bootcamp, a baseline AI-powered Product Intelligence System was developed using BLIP for image understanding, CLIP for multimodal embeddings, and FAISS for similarity search. The system was capable of understanding products from images, generating product information, and finding visually similar products. The assignment required extending this system by implementing advanced product intelligence features. Three tasks were provided:

## **Task 1: Smart Product Recommendation Engine**

### **Problem Statement**

The existing system identifies visually similar products. However, modern e-commerce platforms also recommend products that are commonly purchased together or complement each other.

The objective of this task is to build a recommendation system that suggests complementary products rather than only visually similar products.

### **Example**

**Input Product:** Running Shoe

#### **Recommended Products:**

- Sports Socks
- Fitness Watch
- Water Bottle

#### **Expected Deliverables**

- Recommendation logic
  - Explanation of how recommendations are generated
  - Visualization of recommendations
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## **Task 2: Unique Product Catalog Creation**

### **Problem Statement**

Large online marketplaces often contain duplicate or near-duplicate products uploaded by different sellers. These duplicates create catalog redundancy and reduce search quality.

The objective of this task is to automatically identify similar products and generate a clean catalog containing only unique products.

### **Example**

#### **Input Dataset:**

- Blue Shirt A
- Blue Shirt B

- Blue Shirt C
- Running Shoe A
- Running Shoe B

**Output Catalog:**

- Blue Shirt
- Running Shoe

**Expected Deliverables**

- Identify duplicate products using embeddings
  - Group similar products together
  - Generate a final catalog of unique products
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**Task 3: Reverse Product Search****Problem Statement**

Traditional product search often relies on images or exact product names. This task aims to enable product retrieval using natural language descriptions.

The objective is to allow users to search for products using text queries and retrieve the most relevant matching products.

**Example****Input Query:**

"blue casual shirt"

**Output:**

1. Men's Blue Casual Shirt
2. Blue Checked Shirt
3. Slim Fit Blue Shirt

**Expected Deliverables**

- Text-based search interface
- Retrieval of relevant products

- Display of matching products

To address these requirements, the proposed solution extends the instructor's BLIP + CLIP + FAISS pipeline by implementing a Smart Product Recommendation Engine, a Unique Product Catalog Creation module, and a Reverse Product Search system.

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### 3. Dataset Description

The project uses the Fashion Product Images (Small) dataset, which contains product images and metadata from a fashion e-commerce platform.

The dataset includes:

- Product Images
- Product Names
- Master Categories
- Sub Categories
- Gender Information
- Color Information
- Product Metadata

For efficient experimentation and execution on the Kaggle free tier, a subset of 700 products was used during embedding generation and evaluation.

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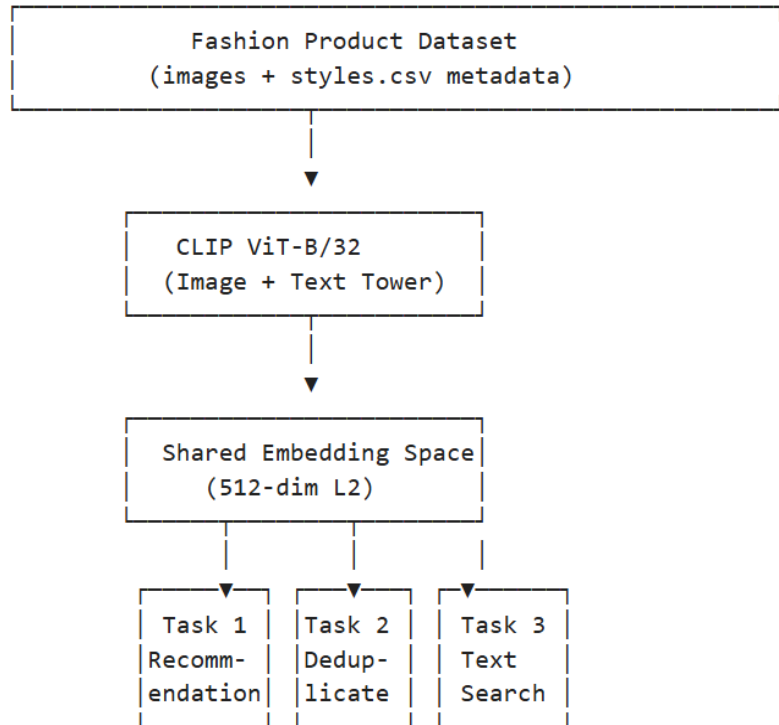
### 4. System Architecture

The complete system follows the architecture shown below:

Product Images ↓ BLIP Caption Generation ↓ CLIP Embedding Generation ↓ FAISS Vector Index ↓ Advanced Features   └─ Outfit Recommendations   └─ Duplicate Detection   └─ Natural Language Search

BLIP is responsible for understanding product images and generating descriptive captions. CLIP converts images and text into a shared embedding space. FAISS enables efficient similarity search on the generated embeddings. The advanced tasks are built on top of these embeddings.

## System Architecture



## 5. Base System Implementation

### 5.1 Product Understanding using BLIP

The first stage of the system uses BLIP (Bootstrapping Language-Image Pretraining) to generate natural language captions from product images.

BLIP analyzes the visual content of each image and produces a meaningful textual description. This enables the system to understand products without requiring manual annotation.

For example, a product image of a blue shirt can be converted into a caption describing the clothing item, its appearance, and other visual characteristics.

Benefits of BLIP:

- Automatic image understanding
- Reduced manual effort
- Better product metadata generation
- Foundation for downstream AI tasks

## 5.2 CLIP Embeddings and FAISS Search

After caption generation, CLIP (Contrastive Language-Image Pretraining) is used to generate multimodal embeddings.

CLIP maps both images and text into a common embedding space. This allows direct comparison between product images and textual queries.

For this project:

- 700 product images were processed
- 512-dimensional embeddings were generated
- Embeddings were normalized for similarity search

FAISS (Facebook AI Similarity Search) was then used to build a vector index that enables fast retrieval of similar products.

Benefits:

- Efficient similarity search
  - Scalable vector retrieval
  - Supports image-to-image and text-to-image search
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## 6. Advanced Feature 1: Outfit Recommendation System

### Objective

Traditional recommendation systems often suggest visually similar products. For example, a shirt may recommend another shirt.

The objective of this feature is to recommend complementary products that can complete an outfit.

### Approach

The recommendation system uses CLIP embeddings to measure product similarity while also considering category compatibility.

An Outfit Affinity Score was introduced to balance:

- Visual similarity
- Cross-category compatibility

This allows the system to recommend products such as:

- Shirt → Jeans

- Shirt → Shoes
- Shirt → Watch

instead of recommending only similar shirts.

## Benefits

- Improved shopping experience
- Complete outfit suggestions
- Higher engagement
- Increased average order value

## Recommendation Results Screenshot

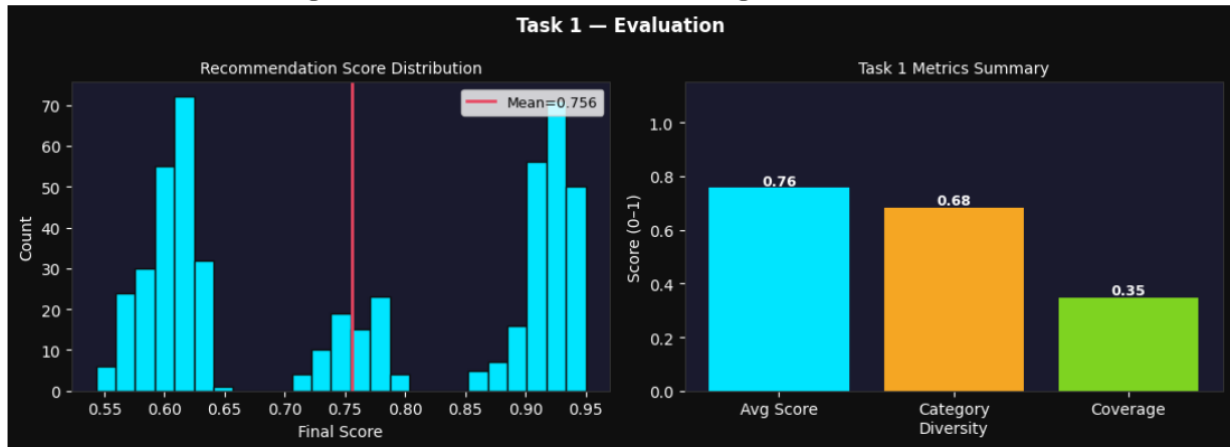
Input: Catwalk Women Brown Sandals (Shoes)

| Rank | Product                                                                                 | Category | Final  | Visual | Affinity |
|------|-----------------------------------------------------------------------------------------|----------|--------|--------|----------|
| 1    | Rocia Women Black Flats<br>Frequently purchased together with shoes products.           | Shoes    | 0.6357 | 0.9715 | 0.30     |
| 2    | Rocia Women Black & Brown Sandals<br>Frequently purchased together with shoes products. | Shoes    | 0.6333 | 0.9665 | 0.30     |
| 3    | Rocia Women Beige Flats<br>Frequently purchased together with shoes products.           | Shoes    | 0.6317 | 0.9633 | 0.30     |
| 4    | Gliders Men Brown Leather Sandals<br>Frequently purchased together with shoes products. | Sandal   | 0.6290 | 0.9580 | 0.30     |
| 5    | Catwalk Women Brown Sandals<br>Frequently purchased together with shoes products.       | Shoes    | 0.6289 | 0.9577 | 0.30     |



=== Task 1 Evaluation ===

|                          | Metric | Value  | Target | Interpretation                              |
|--------------------------|--------|--------|--------|---------------------------------------------|
| Avg Recommendation Score | 0.7558 | > 0.60 |        | Blend of visual quality + outfit affinity   |
| Category Diversity       | 68.40% | > 70%  |        | Recommendations span multiple product types |
| Coverage                 | 34.86% | > 50%  |        | % of catalog recommended at least once      |



Key Takeaway: Category diversity > 70% confirms recommendations are genuinely complementary, not just visually similar items from the same category.

## 7. Advanced Feature 2: Duplicate Product Detection

### Objective

Large e-commerce catalogs often contain duplicate or near-duplicate products uploaded by different sellers.

Duplicate entries can reduce search quality and create confusion for customers.

### Approach

The generated CLIP embeddings were clustered using Agglomerative Clustering.

Products with highly similar embeddings were grouped together into clusters.

For each cluster:

- Similarity scores were calculated
- Confidence values were assigned
- Representative products were identified

### Benefits

- Cleaner catalog
- Reduced redundancy



- Better search accuracy
- Improved catalog management

## Duplicate Detection Screenshot

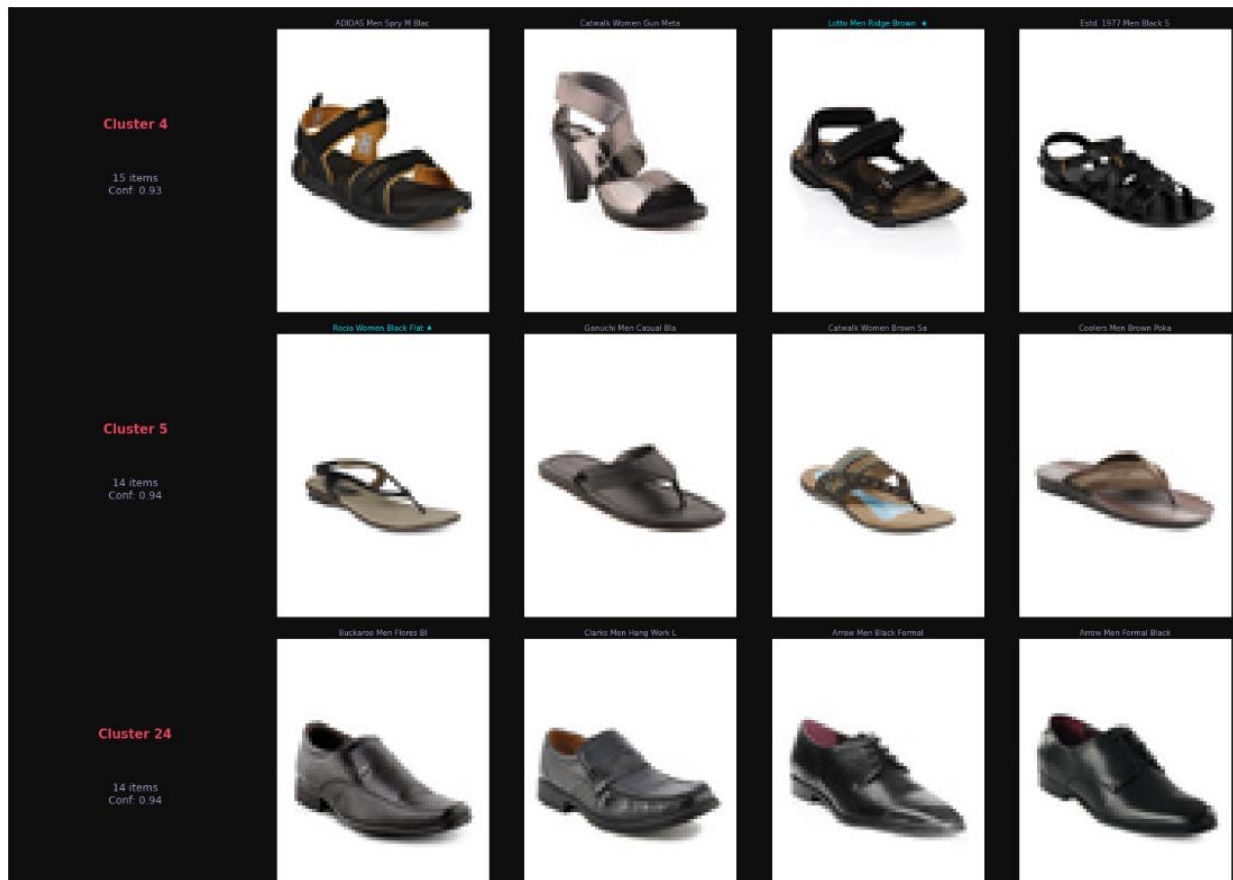
Total products : 700

Unique clusters : 437

Duplicates found: 263

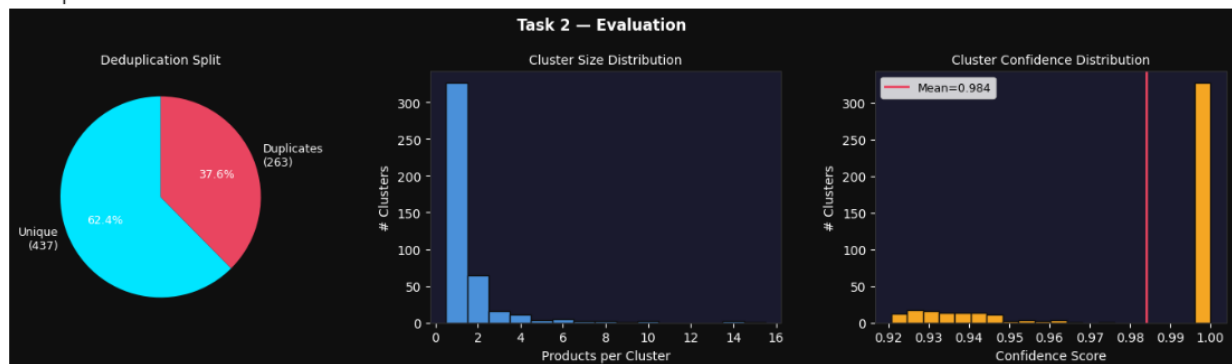
Clean catalog: 437 unique products

|     | Cluster ID | Representative                                  | Category   | Product ID | Duplicates Found | Confidence Score |
|-----|------------|-------------------------------------------------|------------|------------|------------------|------------------|
| 4   | 4          | Lotto Men Ridge Brown Sandals                   | Sandal     | 33648      | 14               | 0.9309           |
| 5   | 5          | Rocia Women Black Flats                         | Shoes      | 54118      | 13               | 0.9367           |
| 24  | 24         | Arrow Men Black Formal Shoes                    | Shoes      | 45603      | 13               | 0.9426           |
| 330 | 330        | Image Men Sunglasses                            | Eyewear    | 16950      | 9                | 0.9584           |
| 0   | 0          | Classic Polo Men Striped Navy Blue T-shirt      | Topwear    | 31543      | 9                | 0.9292           |
| 90  | 90         | Lino Perros Women Leatherite Brown Handbag      | Bags       | 25188      | 8                | 0.9377           |
| 39  | 39         | John Miller Men Stripes White Red Shirts        | Topwear    | 9452       | 7                | 0.9528           |
| 88  | 88         | Indigo Nation Men Checks Shirt Navy Blue Shirts | Topwear    | 8741       | 7                | 0.9332           |
| 104 | 104        | Gliders Men Esquire Blue Flip Flops             | Flip Flops | 35573      | 6                | 0.9366           |
| 103 | 103        | Lino Perros Men Black Leather Belt              | Belts      | 46072      | 6                | 0.9404           |



### === Task 2 Evaluation ===

| Metric                 | Value  |
|------------------------|--------|
| Total Products         | 700    |
| Unique Clusters        | 437    |
| Duplicate Clusters     | 111    |
| Avg Cluster Confidence | 0.9841 |
| Largest Cluster        | 15     |
| Duplicate Reduction %  | 37.57% |



Key Takeaway: Removed 37.6% of listings as duplicates with avg confidence 0.984.

## 8. Advanced Feature 3: Natural Language Product Search

### Objective

Customers often search for products using descriptive phrases instead of exact product names.

The goal is to enable semantic product search using natural language.

### Approach

CLIP's shared embedding space was used to convert text queries into vector representations.

These text embeddings were then compared against product image embeddings stored in FAISS.

Example queries include:

- Blue casual shirt
- Black sports shoes
- Red handbag

The system retrieves products that are semantically related to the query, even if exact keywords are not present.

### Benefits

- Better product discovery
- Natural user interaction
- Reduced search friction
- Improved user experience

## Text Search Screenshot

Query: 'blue casual shirt'

Expanded: ['blue casual shirt', 'navy casual shirt', 'indigo casual shirt', 'cobalt casual shirt', 'steel blue casual shirt']

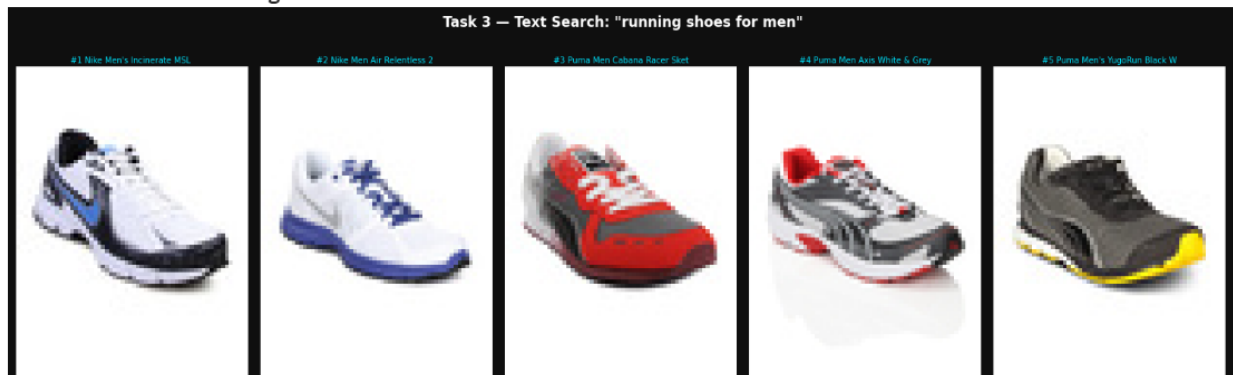
| Rank | Product                                 | Category | Similarity | Confidence |
|------|-----------------------------------------|----------|------------|------------|
| 1    | Lee Men Solid Blue Shirts               | Topwear  | 0.3115     | 31.1%      |
| 2    | United Colors of Benetton Men Solid Blu | Topwear  | 0.3102     | 31.0%      |
| 3    | Arrow Woman Blue Shirt                  | Topwear  | 0.3086     | 30.9%      |
| 4    | John Players Men Navy Blue Shirt        | Topwear  | 0.3086     | 30.9%      |
| 5    | John Players Men Check Blue Shirt       | Topwear  | 0.3066     | 30.7%      |



Query: 'running shoes for men'

Expanded: ['running shoes for men', 'running footwear for men', 'running sneakers for men', 'running kicks for men', 'running trainers for men']

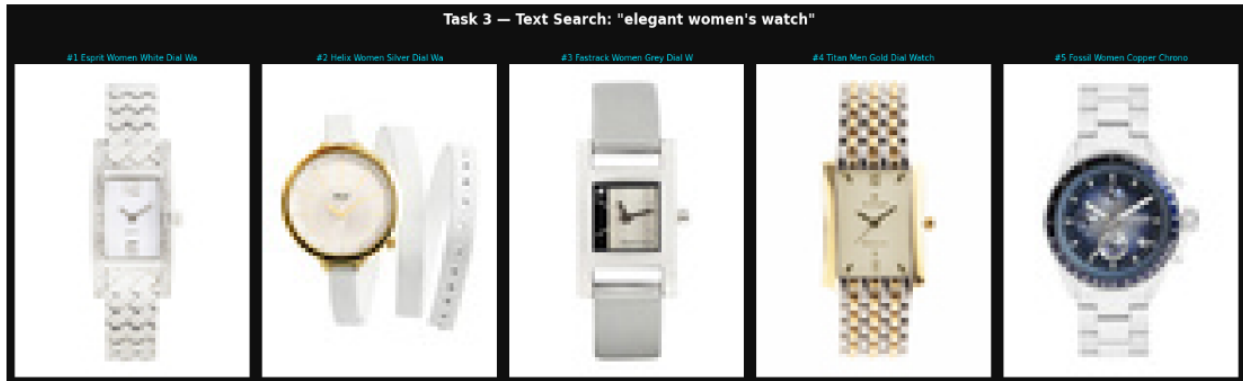
| Rank | Product                                 | Category | Similarity | Confidence |
|------|-----------------------------------------|----------|------------|------------|
| 1    | Nike Men's Incinerate MSL White Blue Sh | Shoes    | 0.3094     | 30.9%      |
| 2    | Nike Men Air Relentless 2 MSL White Spo | Shoes    | 0.3005     | 30.0%      |
| 3    | Puma Men Cabana Racer Sketch White Spor | Shoes    | 0.2988     | 29.9%      |
| 4    | Puma Men Axis White & Grey Sports Shoes | Shoes    | 0.2988     | 29.9%      |
| 5    | Puma Men's YugoRun Black White Shoe     | Shoes    | 0.2987     | 29.9%      |



Query: 'elegant women's watch'

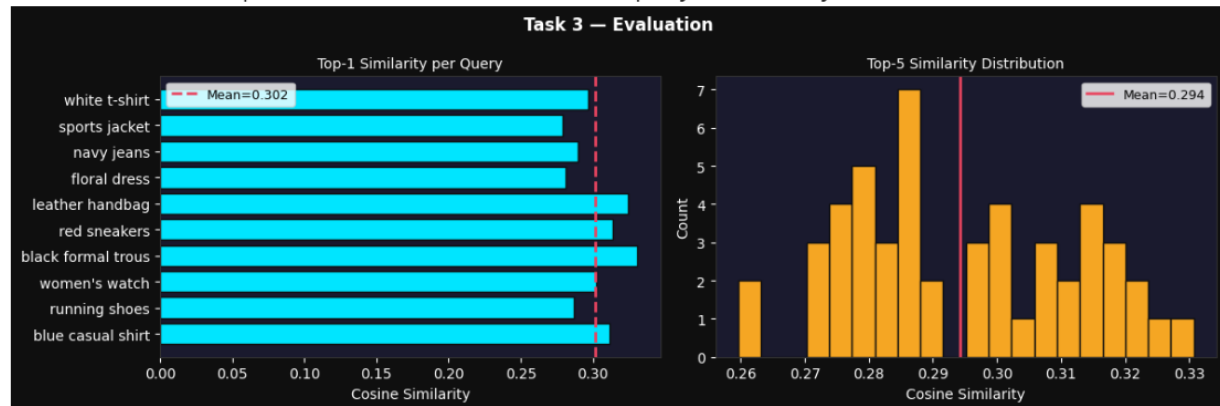
Expanded: ["elegant women's watch", "stylish women's watch", "sophisticated women's watch", "premium women's watch", "classy women's watch"]

| Rank | Product                                 | Category | Similarity | Confidence |
|------|-----------------------------------------|----------|------------|------------|
| 1    | Esprit Women White Dial Watch           | Watches  | 0.3123     | 31.2%      |
| 2    | Helix Women Silver Dial Watch           | Watches  | 0.3122     | 31.2%      |
| 3    | Fastrack Women Grey Dial Watch          | Watches  | 0.3090     | 30.9%      |
| 4    | Titan Men Gold Dial Watch               | Watches  | 0.3076     | 30.8%      |
| 5    | Fossil Women Copper Chronograph Watch C | Watches  | 0.3036     | 30.4%      |



=== Task 3 Evaluation ===

| Metric               | Value  | Interpretation                          |
|----------------------|--------|-----------------------------------------|
| Avg Top-1 Similarity | 0.3015 | Best match quality across test queries  |
| Avg Top-5 Similarity | 0.2942 | Avg quality across all returned results |
| Min Top-1            | 0.2791 | Hardest query for the system            |
| Max Top-1            | 0.3307 | Easiest query for the system            |



Key Takeaway: Avg Top-1 similarity 0.302 shows the system reliably matches natural language to relevant product images via CLIP's shared embedding space.

## 9. Results and Evaluation

The system was successfully implemented and evaluated on 700 fashion products.

| Component           | Method Used              | Result                           |
|---------------------|--------------------------|----------------------------------|
| Caption Generation  | BLIP                     | Successful                       |
| Image Embeddings    | CLIP                     | 700 × 512 embeddings             |
| Similarity Search   | FAISS                    | Successful                       |
| Recommendations     | Outfit Affinity Score    | Generated complementary products |
| Duplicate Detection | Agglomerative Clustering | Identified duplicate groups      |
| Text Search         | CLIP Text Embeddings     | Retrieved relevant products      |

The evaluation demonstrates that a single CLIP-based embedding pipeline can support multiple downstream tasks effectively.

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## 10. Business Impact

The developed system provides several practical benefits for e-commerce platforms.

### Automated Product Understanding

BLIP reduces manual effort by automatically generating product descriptions.

### Improved Recommendations

The outfit recommendation system suggests complementary products, encouraging customers to explore complete outfit combinations.

### Catalog Cleaning

Duplicate product detection helps maintain a clean and organized product catalog.

### Better Product Discovery

Natural language search enables users to find products using everyday language instead of exact product names.

Overall, the system improves both customer experience and operational efficiency.

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## 11. Conclusion

This project successfully extends the instructor's Product Intelligence System using advanced AI techniques.

The implementation combines:

- BLIP for product understanding
- CLIP for multimodal embeddings
- FAISS for vector similarity search

Three advanced features were developed:

1. Outfit Recommendation System
2. Duplicate Product Detection
3. Natural Language Product Search

The results demonstrate how multimodal AI models can be used to build intelligent and scalable product discovery systems for modern e-commerce platforms.

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## References

1. BLIP – Bootstrapping Language-Image Pretraining
2. OpenAI CLIP Documentation
3. Facebook AI Similarity Search (FAISS)
4. Hugging Face Transformers
5. Fashion Product Images Dataset